

Pay for Training, Not for Openness: Reward Design for Original and Distilled AI Models

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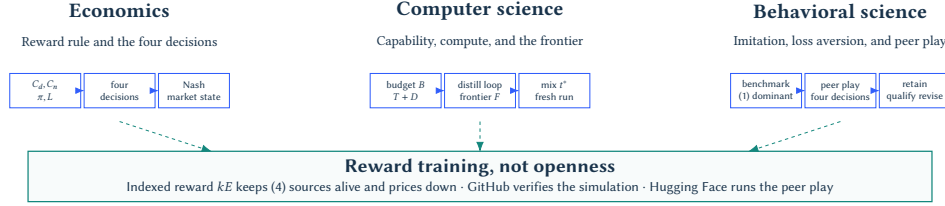


Figure 1: One integrated PS2 evidence chain: economics fixes the four decisions and the reward rule, computation checks whether distillation creates capability, and behavioral evidence shows how firms choose.

Submission to: COMSCI/ECON 206 — Computational Microeconomics.

Course session: Autumn 2026 Session 1. **Team:** FP4.

Course instructor: Professor Luyao Zhang. **Assignment:** PS2 collaborative research proposal.

1 Three connected research questions and verified gap

A firm builds a model by training from scratch at cost C_n , or by distilling an existing model at cost $C_d \ll C_n$ [4, 11]. It keeps the model closed for monopoly profit π , or releases it openly for a reward kE — coefficient k times the training compute E of the release — while losing exclusivity at cost L .

Today’s rules reward *openness*, so a distilled copy and an original source earn the same [6, 25]. The market then settles in state 1 (all closed), state 2 (all distilled: no source, no new capability), or state 3 (healthy: enough sources, reasonable price). Verified gap: no comparison exists between a lump-sum reward for openness and a reward indexed by training compute.

Q1—Economics: Which reward rule aligns the private and social rankings of the four decisions?

Q2—Computation: Does distillation create capability, or only spend compute?

Q3—Behavior: What does each decision pay a firm, and how does it learn which one to choose?

Shared question: can an E -indexed reward hold state 3, at what k , and what share of compute should go to training? Figure 1 links the three answers.

2 Answer to Q1: incentives and strategic model

Players: N firms; actions $\{\text{closed, open}\} \times \{\text{distill, train}\}$; one simultaneous move; common-knowledge parameters with private cost multipliers and market power; objective: own payoff; solution concept: dominant strategies, then Nash equilibrium [8, 10, 21, 24, 28]. Table 1 gives the four decisions.

Table 1: The four decisions, their payoffs, and the reversed rankings (1 = most wanted). An open release also keeps the future market share it buys, $\beta\phi K$: the discounted value of the external contributions an open model attracts [9, 35].

Decision	Payoff	Future	Priv.	Soc.	Reading
(1) closed, distill	$-C_d + \pi$	0	1	4	free rider [23]
(2) open, distill	$-C_d + kE_d - L$	$\beta\phi K$	3	2	passer-on
(3) closed, train	$-C_n + \pi$	0	2	3	monopolist
(4) open, train	$-C_n + kE_n - L$	$\beta\phi K$	4	1	source

The private ranking (1) $\succ$ (3) $\succ$ (2) $\succ$ (4) exactly reverses the social ranking (4) $\succ$ (2) $\succ$ (3) $\succ$ (1): the privately best decision is cheap and exclusive but destroys the source: knowledge is non-rival, so private returns fall short of social returns [1, 26]. Without policy the market stops at (1), a dominant strategy at $k = 0$.

Tool 1 rewards openness only: (2) pays $R - C_d - L$ and (4) pays $R - C_n - L$, so since $C_n \gg C_d$, (2) beats (4) at every R — nobody trains, sources run out, state 2. Tool 2 indexes the reward by training compute, kE : (4) has large E , (2) has small E , and (1) and (3) receive nothing. Table 2 reports the fresh run.

The future-share motive changes where an unregulated market settles. With it switched on, firms release weights with no reward at all, because openness buys a place in the next generation of infrastructure: at $k = 0$ the share of firms opening rises from 0.00 to 0.68 and the market settles in state 2 rather than state 1 — firms open, but what they open is a copy. Opening pays exactly for firms whose share of compatible applications lies between the two roots of $\beta\phi K m / (m + m_0) = L + \pi m$, the inverted-U relation the literature reports [9, 35]. The indexed reward still reaches state 3, and the cap still returns the market to state 2.

The thresholds are analytic. Disclosure needs $kE_d > L + \pi - \beta\phi K$: with the motive off this is $k > (L + \pi)/E_d$ (the familiar $k > L$ assumed $\pi = 0$), and with a strong motive disclosure pays at $k = 0$. Innovation needs $k > \Delta C / \Delta E$, the cost and compute gaps between training and distilling, and this threshold is unaffected by the future-share motive because the term is common to both open cells [20, 34]. Appendix C carries the three lenses.

Table 2: Reward design and market outcome (fresh run, $N = 20000$, seed 206), share of firms per decision.

Reward rule	(1)/(2)/(3)/(4)	State
<i>(a) no future-share motive, $\beta\phi K = 0$</i>		
none, $k = 0$	.96 / .00 / .04 / .00	1 closed
openness, $R = 20$	.11 / .85 / .00 / .04	2 distill
indexed, $k = 4$	.35 / .00 / .01 / .64	3 healthy
indexed, capped, $k = 12$	.42 / .54 / .02 / .03	2 distill
<i>(b) with the future-share motive, $\beta\phi K = 19.98$</i>		
none, $k = 0$	.28 / .68 / .01 / .03	2 distill
openness, $R = 20$	.01 / .95 / .00 / .04	2 distill
indexed, $k = 4$	.03 / .02 / .00 / .95	3 healthy
indexed, capped, $k = 12$	.03 / .93 / .00 / .05	2 distill

3 Answer to Q2: computation, auction comparison, and artifacts

Openness is not only what a weight release produces. A model served through an interface is distilled by whoever queries it, at industrial scale [22], so distillation is itself a degree of openness — and a firm that bans accounts, caps usage, or writes distillation out of its terms is buying exclusivity back. Countermeasures are the mirror image of openness: the harder they bind, the less open the model really is, and a firm can move that quantity without releasing or withholding a single weight. We keep the release move binary — the closed cell is the case where countermeasures fully bind and their cost sits inside L — but the reward is an all-pay auction [2] whose bid is audited training compute, not openness: each firm’s bid is its reported training compute E , every firm pays it because training is sunk whether or not it wins, and the prize goes to the highest verified E (or is split in proportion to E). All-pay auctions dissipate rents, which is why the cap matters [3, 30]; the bid is real compute, not money, and E is verifiable only by audit [27].

Model: budget $B = T + D$ for training T and distillation D ; only training moves the frontier $F(T) = 1 - e^{-T/T_0}$ [12, 14], while distillation copies it with loss $\delta < 1$ and still spends $c_d > 0$. Metrics: decision shares, market state, and social loss $c_e B + \rho \max(0, T - T_0) - vF(T)$, with ρ pricing duplicated effort [5] and v the frontier. Pseudocode: draw each firm’s type, evaluate the four payoffs, take the logit best response [17], aggregate to the market state; seed 206, with parameters and output in Appendix A.

Fresh run (Appendix A gives every parameter). Mutual distillation burns 500 compute units while the frontier falls from 1.000 to 0.774: copying a copy only decays [29]. In the budget account, social loss is lowest at a training share of $t^* = 0.35$ (+1.35), against +2.00 for all distill and +1.77 for all train, so a positive distillation share is efficient and distillation is cheap but never free. Planned: estimate T_0, ρ, δ .

GitHub: PS2-FP4-CapabilityTransfer (course organization)

Notebook: `k_formula.ipynb` and `verify_k.py` in the repository above: the k thresholds, derived and then checked from scratch

Hugging Face: Prize Rate Game: FP4_PS2 (Hugging Face Space)

A0 poster: A0 poster (final): PS2-FP04-CapabilityTransfer-A0-Poster.pptx / .pdf, same question, result, subsidy case and limitations as this paper

4 Answer to Q3: behavior and interdisciplinary integration

Rational benchmark: payoff maximization, with (1) dominant at $k = 0$. What each decision pays the firm and does to the market:

Table 1 gives what each decision pays the firm and does to the market.

Behavioral mechanism: firms imitate the cheapest visible profitable action, so imitation pushes the market from (1) toward (2); loss aversion makes the lost exclusivity L weigh more than an equal reward kE [13], so firms under-open. Competing explanation: firms release open weights for reputation, hiring, standards and future market share, so openness is cheaper than a model without the $\beta\phi K$ term assumes [7, 9, 15, 33]. Distinguishing observation: hold total reward spending fixed and switch from lump sum to E -indexed — if imitation dominates, the share of (4) rises slowly; under reputation, the switch changes little. The Hugging Face game runs this switch as peer play; if imitation dominates, we retain the model but qualify common knowledge.

5 Advanced development, real-world impact, and roadmap

Foundations: equilibrium reasoning [10, 21, 28] (Nobel 1994), mechanism design [16, 18, 19, 32] (Nobel 1996, 2007), and knowledge distillation [11] (Turing Award 2018). Distinctive now: weights are copied at almost no cost while training stays expensive, and no rule separates the two.

Real-world case: an agency paying for open-weight models, such as a national compute subsidy. Stakeholders: labs, small firms, users, taxpayers, the agency. Benefits: more sources, lower prices, wider access. Risks: paying for copies, crowding out training, gaming the reward. What counts is audited training compute, not self-declared openness [27]; value is original-source releases per unit of spending; the next discriminating test is the reward switch with real firms; the cumulative map is Appendix E.

Expected 2056 Nobel Prize and Turing Award contribution statement. By 2056, we aspire to a reward standard for how a society pays for machine intelligence; progress will be shown by shifts in original-source releases.

Open Science Statement

GitHub: `code/distill_sim.py`, seed 206, log in `code/fresh_run_output.txt`; Hugging Face: Prize Rate Game. Both use the same payoffs and states as this paper.

Statement of Contribution to the UN’s SDGs

Intended contribution to SDG 9.5 and SDG 10: a rule that keeps sources alive widens access, measured by original-source releases per unit of spending [31]. Paying for copies would raise spending without new capability, so the rule must be evaluated first.

Submission milestones

Review-ready September 27, 2026, 11:00 P.M.; symposium September 28, 2:45–5:15 P.M., IB 2050; final revision October 6, 2026, 11:00 P.M. Beijing time.

Author Notes

Acknowledgements

Course instructor: Professor Luyao Zhang. We thank her for incubating this research from its earliest strategic question, for mentoring the interdisciplinary development of the project, and for her formal review of PS1. We thank Prof. Ken Rogerson and Han Zhang (founder of Liberata; Ph.D. candidate at Duke University) for their discussions and feedback; **【填】此处填 peer reviewers 的全名（分组：张老师课堂反馈、同学评审、Han Zhang）** for their reviews; and our COMSCI/ECON 206 classmates for collaborative learning and questions. We acknowledge the course Overleaf template, Python and NumPy. No external model or compute data was used; every number in this paper comes from code/distill_sim.py. The authors remain responsible for the proposal and remaining errors.

Team and individual contributions

Joint contribution. The team asked one question: does rewarding openness buy capability, or only copies? Together we built the four-decision model, defined the two reward tools, and fixed the three market states. We specified a compute account in which only original training moves the capability frontier, and designed the peer-play switch between a lump-sum and an E -indexed reward. The simulation, the state classification, the poster claim, and Appendix D use the same payoffs, parameters, and seed, so every number can be rerun from code/distill_sim.py at seed 206.

Zhengjun He, author order 1. Contribution: the initial idea that rewarding openness pays copies as well as sources, and the Hugging Face peer-play space. Output: Sections 1 and 4; Prize Rate Game: FP4-PS2 (Hugging Face Space).

Yichen Shen, author order 2. Contribution: the three connected questions and the GitHub artifact. Output: Sections 1–3, code/distill_sim.py; PS2-FP4-CapabilityTransfer (course organization).

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A Technical details and reproducibility

Four payoffs per firm: (1) $-C_d + \pi$, (2) $-C_d + kE_d - L$, (3) $-C_n + \pi$, (4) $-C_n + kE_n - L$. Firm types: a simulated population of $N = 20000$ firms (no firm-level data enters; the population only aggregates individual choices), cost multiplier a and market power m drawn independently lognormal with $\sigma_a = \sigma_m = 0.6$, seed 206, NumPy 2.4.6. Choices use a logit best response with $\lambda = 1$ [17]. Level

constants, calibrated so that the run reproduces column (a) of Table 2 (residual 7.4×10^{-5}): $C_d = 1.4446$, $C_n = 5.6704$, $\pi = 8.2391$, $L = 2.5193$, $E_d = 0.9854$, $E_n = 4.6311$. Future-share motive: $\beta = 0.90$, $\phi = 0.50$, $K = 44.4$, $m_0 = 0.50$, giving $\beta\phi K = 19.98$ [9]. Reward designs: (A) lump sum R for any open release; (B) reward kE ; (C) reward $\min(kE, 12)$. Compute account: $F(T) = 1 - e^{-T/T_0}$, mutual-distillation loop with $\delta = 0.95$, $c_d = 1$, 100 firms, 5 rounds, and social loss $c_e B + \rho \max(0, T - T_0) - vF(T)$ with $B = 100$, $T_0 = 25$, $\rho = 0.01$, $v = 1$, $c_e = 0.02$. Outputs: `code/fresh_run_output.txt`. Shortest reproducible run: `python code/distill_sim.py` (in this package). The repository holds `k_formula.ipynb`, which derives the two thresholds on k with sympy, and `verify_k.py`, which recomputes the four cells from scratch and finds where the best cell changes. Results are deterministic. Known limits: the frontier, duplication price, and δ are illustrative, not estimated; firms are symmetric except for a and m ; no entry or exit; no data on real training compute. Repository status: PS2-FP4-CapabilityTransfer (course organization), release commit 383092e.

A.1 AI-use disclosure

Generative AI (DeepSeek) was used only for typesetting, for debugging `distill_sim.py`, and for locating references. The ideas, the model, and every test and number come from the authors, who reran the code at seed 206 and checked each value against Table 2.

B Cumulative Intellectual Development

We started from one observation: a firm can imitate an existing model cheaply or innovate at full cost, and can then keep the result secret or disclose it. A public prize rewards disclosure, so it pays the imitator and the innovator the same amount even though only the innovator paid for the capability. Writing the two choices as a 2×2 table made the conflict visible: the privately best cell (imitate, secrecy) is the socially worst, and the socially best cell (innovate, disclose) is the privately worst. The next question was what the prize should be paid for. Paid per unit of training budget E , it separates the two open cells, so the innovator collects the largest prize and the imitator the smallest; paid flat, the two cells are identical and nobody has a reason to innovate. The last question was whether every firm should innovate. A compute account shows that mutual imitation spends compute without adding capability, while everyone training duplicates the same work, so a mixture is efficient. The GitHub notebook derives the range of the prize rate that keeps the market healthy, and the Hugging Face game tests the first two steps with human players.

Table 3: From foundations to a collaborative interdisciplinary contribution.

Horizon	Milestone and evidence
Foundation	Dominant strategies; reversed private and social orders.
PS2	Reward-rule model, fresh run, peer play, threshold k .
Symposium	A0-poster claim, subsidy case, challenge, response.
Next study	Estimate T_0 , ρ , δ ; field test of the switch.

C Open-source and three-lens inspiration

Sources and tools: the course Overleaf PS2 template (acmart class, ACM license terms, used as the paper skeleton); Python 3.11 with

NumPy 2.4.6 (BSD license, used for the simulation); the references in this paper (published articles, used for concepts only). Game theory changes the model by giving a dominant strategy and a threshold on k . Social choice changes it by showing the private and social orders are exactly reversed, which is why the outcome is not efficient without policy. Mechanism design changes it by turning the reward into an instrument: the designer picks k and the index, and the index matters more than the level. Observed: the simulation shares and the mutual-distillation decay. Interpreted: the three market states. Planned: estimating T_0 , ρ , and δ .

C.1 Evidence from practice, as of 2026-10-03

We compiled an eight-family evidence table (OpenAI, Anthropic, Google DeepMind, Meta, DeepSeek, Alibaba, Mistral AI, xAI; released as `EVIDENCE_TABLE.md` with the repository) recording, for each family, whether weights are downloadable, under which licence, whether training compute is disclosed, and whether distillation from another developer’s model is documented. Three findings matter for this paper. First, openness is a licence spectrum rather than a binary, and most families occupy several of the four decisions at once — Google ships Apache-2.0 weights beside API-only flagships, Meta ships closed frontier models beside Apache-2.0 open weights — which is why a prize indexed on openness alone cannot separate (2) from (4). Second, disclosure of training compute, the quantity our prize is indexed on, is sporadic: no family in the sample states a training-compute figure for its flagship release, while one family reported GPU-hours for a single generation and another reported FLOP for a single model. Third, the best-documented distillation case is an open release distilled from named third-party checkpoints, and a joint NSA/CISA/FBI advisory (AA26-251A, 8 September 2026) [22] alleges industrial-scale distillation by several developers; the allegation remains contested and was officially rejected by MOFCOM. None of this is causal evidence: the table is a dated cross-section, it cannot measure a firm’s performance edge at release, and several cells are documented negatives rather than proofs of absence. It does support the design choice, because audited compute is the only quantity in the table that a rule could verify.

D Review response and revision record

Each substantive comment appears once, with its source, our decision, the change made, where it can be checked, and what still needs testing. Decisions are **Keep**, **Revise**, or **Investigate**. Review-ready v1: September 27, 2026. Reviews and discussant feedback: September 28, 2026 symposium, IB 2050. Final v2: October 6, 2026.

D.1 Instructor, immediate feedback in class

【填】整栏留空：此处填张老师课堂即时反馈。每条给出：意见原文、我们的决定（**Keep / Revise / Investigate**）与理由、改了什么、证据位置（PDF 页码 + 章节）、遗留问题（标 **planned**）。

D.2 Peer reviews received at the symposium

【填】整栏留空：此处填同学评审。每位 **reviewer** 一块，写全名、意见原文、我们的决定与理由、改了什么、证据位置、遗留问题。

D.3 Discussion with Han Zhang (Liberata; Duke Ph.D. candidate)

【填】整栏留空：此处填 **Han Zhang** 的反馈。每条给出意见原文（例如关于市场份额的那一条）、我们的决定与理由、改了什么（模型新增 $\beta\phi K$ 未来市场份额项）、证据位置、遗留问题。

D.4 Reviews we submitted (individual work, two per student)

【填】整栏留空：此处填我们各自提交的两份评审记录（评了谁、要点、后续跟进）。

E Structured abstract and cumulative proposal map

The eight-row cumulative map uses the same question, assumptions, and evidence status as the main paper, the artifacts, and the poster.

Table 4: Appendix E structured abstract; entries match Sections 1–5.

Proposal element	Current statement / change
Background	Firms choose between cheap distillation (C_d) and expensive original training (C_n), and between closed exclusivity (π) and open release (reward kE minus lost exclusivity L). Current rules reward openness only.
Motivation and application	Cheap copying and expensive training coexist, so a rule that pays for copies can empty the market of sources. Case: an agency that pays for open-weight models; affected: labs, small firms, users, taxpayers, the agency.
Three connected questions	Q1 which reward rule aligns the private and social rankings; Q2 whether distillation creates capability; Q3 what each decision pays a firm and how firms learn. Integrated question: can an E -indexed reward hold state 3, and at what k ?
Method and model assumptions	$N = 20000$ firms, one simultaneous move, common-knowledge parameters, private cost multiplier and market power, logit best response, dominant strategy and Nash equilibrium, compute account $F(T) = 1 - e^{-T/T_0}$, seed 206.
Results or expected results	Actual: (1) is dominant at $k = 0$; openness-only rewards never lift sources above 0.04; E -indexed rewards hold state 3 from $k \geq 3$; mutual distillation spends 500 units for no new capability; $t^* = 0.35$. Planned: estimated T_0 , ρ , δ ; a per-firm cap test. Falsifier: an openness-only rule that keeps sources above 0.10 in a rerun.
Intellectual merits	Beyond prior work, the comparison is between two reward rules over the same four decisions, and the reward index (openness versus training compute) is shown to matter more than its level.
Practical impacts	Private value: profit and reward. Public value: sources alive, lower prices, wider access (SDG 9.5, SDG 10, intended only). Risks: paying for copies, crowding out training, gaming. Evidence needed: share of original-source releases per unit of spending. Correction route: published rules and audits.
Feedback received and revision made	The capped-reward run changed the recommendation: with a flat or capped prize the differential disappears and the market returns to state 2, which added the cap warning in Section 2 and the falsifier “an openness-only rule that keeps sources above 0.10”.

The main paper, the artifacts, the poster, and this map use the same question, payoffs, parameters, and evidence status.

F PS2 extension record: mechanism, linked artifacts, poster, and symposium

F.1 Auction laboratory record

The reward rule is an all-pay auction [2]. Scarce resource: the annual public reward budget. Participants: firms that release models. Bid: the verified training compute E of the release. Payment rule: every firm pays its bid, because the training cost is sunk whether or not it wins. Allocation rule: the highest verified E takes the prize; if the budget is shared, each firm receives a share proportional to $E/E_{\max}$. Stopping rule: the budget is exhausted. Predicted equilibrium: over-dissipation, so bids rise above the socially optimal training level until the cap binds — the capped row of Table 2. Treatments and the values to be recorded after the September 28 play:

Treatment	Predicted equilibrium	Observed, utility, revision
Lump sum R	(1) or (2); no sources; state 2	[record share, payoff, revision]
Indexed kE	(4) modal above $k = 3$; state 3	[record share, payoff, revision]
Indexed, capped	Back to state 2 at large k	[record share, payoff, revision]

Allocation efficiency is measured as the share of firms choosing (4); revenue is the public spending per unit of new capability.

F.2 Linked-artifact parity

Artifact	Required parity check
GitHub	Same four payoffs, parameters, seed 206, runnable paths <code>k_formula.ipynb</code> , <code>verify_k.py</code> and <code>code/distill_sim.py</code> , and the limits stated in Appendix A.
Hugging Face	Prize Rate Game: same decision setting, same payoffs per round, same evidence status, and no personal data from peer play.
A0 poster	Same question, same state-3 claim, same subsidy case, same limitations, same instructor and links.

F.3 Symposium and practical-impact record

[September 28, 2026 symposium: presentation version, questions received, and the team’s decisions. For the subsidy case: stakeholders, proposed benefit, risk, missing evidence, correction route. Then record the changes made for the September 30 final revision and rerun the affected check.]